Cooperative Lunar Exploration with ROS 2: Energy, Communication and Recoverable Task Allocation

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***Abstract*: This work integrates two simulated rovers, visual perception, energy management and recoverable task allocation for cooperative lunar exploration. ROS 2 Jazzy nodes communicate with Gazebo Harmonic through typed bridges. Adapted Space ROS plugins provide rechargeable batteries and solar generation with rendered occlusion, while a mission gateway introduces repeatable communication faults. Rovers share camera discoveries, inspect allocated sites and return to separate bases. A nearest-rover baseline and an energy-aware heuristic use the same feasibility and navigation rules. Actual Gazebo validation completes a compact mission and an 80 by 60 m obstacle-rich failure scenario, with five confirmations, both rovers returned and no conservative footprint overlaps. All 16 matched policy runs complete, with effects varying between scenarios. The system demonstrates package integration and observable failure recovery; its accelerated power model, known terrain and ground-truth localisation constrain conclusions about real lunar operation.**

***Index Terms*: multi-robot systems, ROS 2, Gazebo, task allocation, energy, fault recovery**

I. Introduction

Exploration requires more than sending independent robots to waypoints. A rover may discover a site that another can inspect with less travel or greater remaining energy. A temporary failure can leave an assignment unfinished. The team needs shared observations, explicit ownership, resource-aware decisions and a defined completion condition. Multi-robot task allocation offers a framework for distinguishing task requirements and robot resources [1].

Our RSDIS project investigates these concerns in a lunar-inspired mission. Two rovers survey different sectors, discover five visual markers, exchange their estimated locations and allocate inspections. Success requires all five sites to be confirmed and both vehicles to return to their own bases. The research question is whether adding battery risk to a greedy distance score changes completion, travel and energy use under identical safety constraints.

The engineering contribution is the integration of existing simulation, middleware and perception facilities into a reproducible cooperative system. The group develops the mission agents, task protocol, perception pipeline, navigation safeguards, power governor, communication gateway and observers. Existing models, plugins and standard algorithms are credited explicitly. We do not claim a novel optimal allocator or flight-qualified lunar autonomy.

II. Reused Components and Integration

The runtime targets Ubuntu 24.04, ROS 2 Jazzy and Gazebo Harmonic. Stock Gazebo systems provide physics, differential drive, joint control, cameras, lidar and ground-truth odometry. The installed `ros_gz` bridge translates sensor, command and power messages between Gazebo Transport and ROS 2 [2]. OpenCV supplies image conversion, HSV segmentation and contour operations [3]. These dependencies provide infrastructure rather than the cooperative mission itself.

Space ROS supplies a lunar exploration demonstration and simulation repository [4], [5]. We pin these at revisions `4b4eb8447423` and `13226b067d92`, respectively. The default world uses lightweight rover geometry for practical software-rendered experiments. An optional adapter retains the upstream VIPER-inspired meshes, inertias and steerable joints, with namespaced controllers and steering-feedback interlocks. Upstream terrain assets are available separately; the evaluated outpost is our generated planar scene.

The energy package adapts five Space ROS C++ source/header files implementing rechargeable batteries and solar panels. Changes include per-rover power interfaces, held inputs with watchdogs, ROS battery units, exhaustion recovery and rendering-thread handling. Original Apache-2.0 notices remain in the files. A manifest records immutable upstream URLs and original/adapted SHA256 hashes; an automated check verifies attribution. The source guide distinguishes copied code, installed dependencies and group-authored logic.

III. Mission Architecture

Figure 1 shows the runtime boundaries. Each rover has an agent, camera detector and power governor. A shared coordinator allocates inspections. A gateway mediates mission traffic, while local sensors and motion control remain available during a mission-link outage. Independent observers

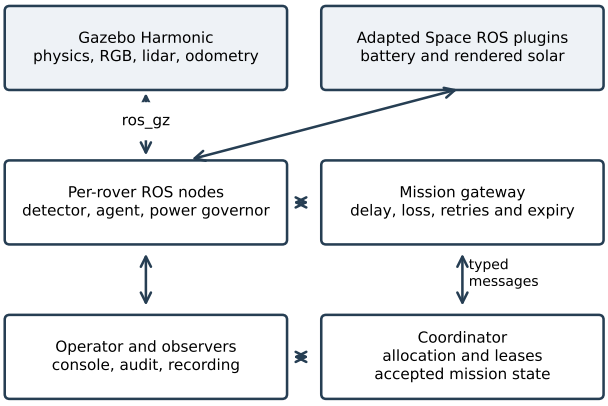


Fig. 1. Runtime integration. Grey blocks contain existing simulation facilities; white blocks contain project ROS logic. Observer truth is never fed into target detection or task allocation.

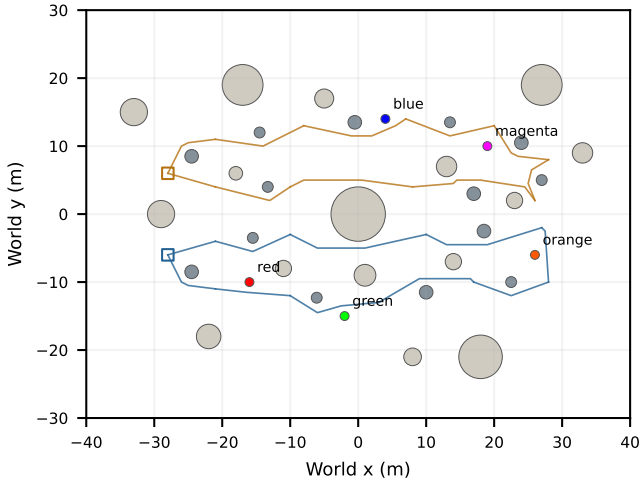


Fig. 2. Expanded outpost and illustrative planned survey routes. Darker rocks are the added barriers. Paths illustrate A* detours; target locations are shown for map design, but are discovered from images at runtime.

collect geometry audits, electrical telemetry, camera error and outcomes.

A. Interfaces and State Ownership

Custom messages represent detections, rover status, leased goals, results and authoritative mission state. Sensor and actuator topics use `/rover1` and `/rover2` namespaces; shared topics use `/mission`. All mission positions refer to a common world frame. The coordinator alone decides whether a result belongs to the current owner and task identifier. It publishes accepted results and a complete latched snapshot, preventing a delayed obsolete result from making a site appear confirmed in the console.

Local images, lidar, odometry and status use sensor-style best-effort delivery. Goals, detections and results use reliable mission interfaces. Authoritative state uses reliable, transient-local delivery with depth one, so a late observer can reconstruct the current state. These are existing ROS 2 DDS QoS facilities [6]; application leases provide a separate freshness contract.

The local web console shows actual RGB frames, observed sites, planned and travelled routes, battery voltage/current and source power. Operators can pause, resume, recall, disable a rover or disconnect its mission link. Recorded checkpoints are labelled and their control buttons disabled. Pausing stops mission commands while Gazebo and native battery physics continue.

B. Discovery and Local Confirmation

Five uniquely coloured spheres stand in for scientific sites. The detector thresholds HSV masks and rejects small, clipped or strongly non-circular contours. Known marker diameter and camera calibration give approximate range. If D is the diameter, α half the angular separation of the horizontal silhouette edges and β their mean bearing, the level-camera estimate is

$$z_c \simeq \frac{D\cos\beta}{2\sin\alpha}, \qquad y_c \simeq -\frac{D\sin\beta}{2\sin\alpha}. \tag{1}$$

Camera extrinsics and the closest timestamped rover pose project the estimate into world coordinates. Images without a pose within 0.2 s are discarded. Unique colour IDs suppress duplicate target records; inconsistent position updates are rejected. This is known-size marker ranging, not mineral recognition.

An inspection requires reaching a stand-off, facing the marker and receiving recent local camera observations over a dwell interval. A teammate's report can create an assignment but cannot alone confirm it. The detector does not load target coordinates. A separate supervisor compares delivered estimates with simulator truth solely for evaluation.

IV. Map and Motion Control

The expanded outpost covers 80 by 60 m under lunar gravity. It contains 24 rocks, four crater exclusions, a habitat and a solar array. Fourteen added rocks create detours between survey stops without enclosing scientific sites or blocking charging pads. All 20 prescribed survey legs remain connected for conservative rover radii of 0.72 m and 1.35 m; 18 require deviations from the direct segment. Figure 2 depicts design geometry, not a recorded trajectory. A 40 by 30 m compact field supports faster matched experiments.

A bounded planar A* planner [7] inflates prior hazards by vehicle footprint and safety margin. Camera-derived marker estimates and peer positions also form exclusions. Continuous edge checks prevent grid connections through obstacles. The agent aligns before translating and evaluates a sampled curved sweep over reaction and braking time before issuing a command.

A front-mounted, 180-degree, 721-ray lidar operates at 10 Hz. Valid endpoints provide eight seconds of local obstacle memory; peer returns do not leave static trails. Missing or invalid scans stop motion. Facing rovers commit to right-side passing points to avoid symmetric oscillation. Inspection approaches remain fixed while the vehicle turns, avoiding pursuit of a shifting noisy estimate. Survey stops perform camera sweeps to discover sites outside the forward view.

Known terrain and ground-truth pose isolate allocation from localisation uncertainty. Craters are visual depressions with navigation exclusions rather than a physical heightmap. The conservative circular footprint audit reports geometric separation, not contact forces or traversability of lunar soil.

V. Energy and Task Allocation

A. Native Simulator Energy

Gazebo supports battery state and consumer power as simulation entities [8]. Our adapted rechargeable plugin receives electrical load from a ROS governor using measured odometry:

$$P_L = 10 + 100\min(|v|, 1) + 15\min(|\omega|, 2) \quad \text{W}. \tag{2}$$

The model uses 24 V nominal voltage and 0.25 Ah capacity. Solar supply is $P_S = 12\max(0, \mathbf{n}\cdot(-\mathbf{s}))$ W when an outward ray from the panel towards the directional sun is unoccluded, and zero otherwise. Scene updates perform the query in the rendering thread. A stationary rover within 0.3 m of its own pad may receive 550 W dock power in an appropriate charging state. These accelerated parameters make charging observable during a short demonstration; they are not calibrated VIPER specifications.

Ignoring clipping and transient smoothing, charge evolves as

$$\dot{q} \simeq \frac{P_S + P_D - P_L}{3600V}, \qquad b = q/C. \tag{3}$$

The plugin bounds q within $[0, C]$, publishes at 10 Hz simulation time and expires a held source after one second without updates. ROS BatteryState uses amperes, ampere-hours, a fraction in $[0, 1]$ and negative discharge current [9]. The bridge copies these fields directly; adapting the upstream percentage and current convention is therefore necessary. Stale battery data makes the agent stop and become unavailable rather than silently switching models. The portable rehearsal retains an explicitly labelled synthetic approximation.

B. Shared Feasibility and Different Scores

The coordinator stores at most one owner per target and one active inspection per rover. Let d_{ij} be approximate outward distance to the stand-off, h_{ij} estimated return distance, b_i reported battery fraction and $r = 0.15$ reserve. Its simple prediction is

$$e_{ij} = k_d(d_{ij} + h_{ij}) + k_t\left((d_{ij} + h_{ij})/v_{\text{ref}} + 10\right). \tag{4}$$

Both policies reject a candidate if $b_i < e_{ij} + r$. The nearest-rover baseline minimises d_{ij}; the energy-aware heuristic minimises

$$J_{ij} = d_{ij} + \lambda\frac{e_{ij} + r}{\max(b_i, 0.01)}. \tag{5}$$

Parameters are $k_d = 0.006$ per metre, $k_t = 0.0001$ per second, $v_{\text{ref}} = 0.3$ m/s and $\lambda = 8$ m. Target-ID order and rover-ID tie-breaking make the greedy decision reproducible. The predictor is a resource-budget heuristic, distinct from the measured electrical model. It is not fitted online and cannot account exactly for A* detours, rotations or solar occlusion. Agents also check their return budget and can release work for recharging.

C. Recovery and Communication Faults

Goals carry task IDs and two-second renewable leases. Unavailable or returning rovers, stale heartbeats and task timeouts release ownership. Retried targets receive a new task ID after a cooldown, with at most three attempts. Duplicates and results for old assignments are ignored. Lease expiry is checked against the original timestamp, so delayed delivery cannot extend an obsolete task. A late heartbeat cannot replace a newer pose.

The gateway supports ideal delivery and a seeded degraded profile: 15% loss per attempt, 0.12 s delay and uniform ± 0.05 s jitter. A rover-1 blackout from world time 20 to 32 s exercises loss of mission connectivity. Reliable application traffic retries every 0.2 s until expiry; best-effort heartbeats are not accumulated for later delivery. Goal renewals are coalesced and the queue is bounded at 512 entries. Independent random streams per rover and channel avoid coupling unrelated traffic through one generator.

This is an application-channel fault emulator above ROS, not a radio propagation model or measurement of DDS packet loss. Local sensing remains active, and operator pause, recall and final-stop channels are out-of-band. The experiment therefore evaluates selected mission-link faults, not a complete partition.

VI. Evaluation

A. Reproducible Design and Measurements

The frozen comparison contains nominal, asymmetric-energy, temporary-failure and communication-outage scenarios. Two seeds perturb target x, y coordinates by at most 0.2 m before collecting outcomes. Each configuration is executed under both policies, giving eight matched layouts and 16 Gazebo runs. Nominal batteries start at 0.95 and 0.55; the asymmetric case uses 0.30 and 0.95. Failure disables rover 1 at fixed world time 30 s for 30 s under both policies. The separate expanded-map functional gate can trigger failure on an active task; it is not pooled with matched comparisons. Each run has a 900 s simulated deadline and uses native energy.

Observers retain success including return, confirmations, elapsed simulation time, odometry distance, minimum geometric gap and overlap samples. Electrical discharge is integrated as $\int \max(0, -VI)dt/3600$ Wh over the mission window;

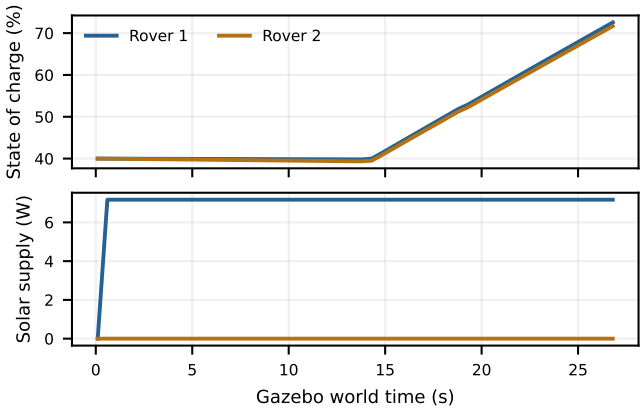


Fig. 3. Actual native-energy bench, revision `6351d85`. Rover 1 is exposed to sunlight and rover 2 is under a rendered canopy. Dock recharge follows the initial discharge interval. Capacity and dock power accelerate the test.

solar and dock inputs are recorded separately so charging cannot cancel prior discharge. Gateway counters record attempts, losses, retries, expiries and delivery delay. Camera-location error is evaluated against truth outside the control loop. Failed runs remain in the dataset.

Each run stores configuration, CSV telemetry, JSONL events, source fingerprint, package versions, intervention service outcomes and termination reason. The recorder combines genuine Gazebo overhead/RGB images with ROS telemetry and can save a ROS bag. It does not generate a mission from oracle detections. Software rendering makes wall time host-dependent; comparisons use simulation time. With only two seeds per scenario, results are descriptive rather than evidence of statistical superiority.

B. Functional Validation

Revision `6351d85` passes Jazzy/Harmonic integration, including physical rotation in both directions, translation, lidar, actual cameras and task messages. The compact mission finishes in 235.0 simulated seconds. The expanded obstacle/failure mission confirms all five sites and returns both rovers in 746.5 s, travelling 423.4 m with minimum conservative obstacle clearance 0.2824 m and zero overlap samples. Both failure service calls succeed. Red-site ownership transfers from rover 1 to rover 2, which confirms it; this is actual cross-rover recovery rather than an offline rehearsal.

The native energy bench verifies exposed-panel production, rendered shade, discharge, dock recharge and ROS units. Figure 3 shows recorded electrical telemetry. The exposed panel supplies approximately 7.17 W under the configured sun incidence; the covered panel supplies zero. The strengthened bench at revision `6223585` also verifies discharge to zero and recovery. The detailed VIPER profile passes its own native-energy compact gate in 157.0 s, travelling 72.0 m with minimum obstacle gap 0.5222 m and zero overlap samples. This gate does not independently establish VIPER solar occlusion or performance on expanded or uneven terrain.

TABLE I
ACTUAL PAIRED GAZEBO RESULTS. EACH ROW AVERAGES TWO LAYOUTS. N: NEAREST ROVER; E: ENERGY-AWARE. ENERGY IS CUMULATIVE BATTERY DISCHARGE.

Scenario	Policy	Done	Time (s)	Dist. (m)	Wh
Nominal	N	2/2	126.7	54.9	1.785
Nominal	E	2/2	107.8	53.2	1.695
Asymmetric	N	2/2	116.0	48.7	1.588
Asymmetric	E	2/2	116.5	48.1	1.574
Rover failure	N	2/2	141.2	59.3	1.927
Rover failure	E	2/2	125.0	49.4	1.626
Link outage	N	2/2	139.8	55.8	1.875
Link outage	E	2/2	137.0	56.5	1.872

C. Paired Policy Outcomes

The complete batch uses runtime `6223585` and one verified source fingerprint. Table I reports the mean of two predefined layouts per scenario and policy; individual outcomes are retained in the repository. No failed run is removed.

16 of 16 runs complete with five confirmations and return. The minimum conservative obstacle gap across the batch is 0.5331 m, with 0 overlap samples. Both failure interventions are valid in each failure case. Across the eight matched layouts, the mean E-minus-N difference is -9.37 s, -2.90 m and -0.102 Wh. These differences describe this small batch. Variation between layouts and ROS/Gazebo execution schedules prevents attributing every timing difference solely to the heuristic. There are no repeated executions of each fixed case, and these data do not establish statistical superiority.

Delivered-detection position RMSE ranges from 0.190 to 0.328 m per run. Repeated frames are correlated and sample counts differ with the routes; this is a descriptive perception check. The outage cases record 3052 lost application delivery attempts and 460 retries across both policies, while all four finish.

VII. DISCUSSION AND LIMITATIONS

The system makes the relationship between resources, observations and task ownership inspectable. Shared discovery permits one rover to inspect another's observation. Freshness checks and accepted state address the distinction between receiving a message and accepting a current result. The power bench tests the bridge independently of allocation, and the expanded map tests navigation and recovery independently of the compact policy comparison.

Several limits remain. Ground-truth pose removes localisation drift, the prior map contains terrain exclusions, and coloured markers simplify perception. Planar control and circular audits do not evaluate soil mechanics, slopes or contact forces. Battery coefficients, capacity and dock power are demonstration parameters; the fraction-based predictor is not a physical route-energy optimiser. The central coordinator is a single point of failure. Seeded channel faults do not replace hardware network measurements. Two small layout perturbations cannot establish generalisation across lunar terrain or prove that either allocator is globally better.

Useful extensions are measured route-energy prediction including detours, more independent layouts, onboard localisation, uneven terrain and distributed coordination. Each should preserve the separation between observer truth, operational measurements and declared assumptions.

VIII. Conclusion

The project combines existing Space ROS plugins, Gazebo systems, ROS 2 QoS and OpenCV with custom cooperative mission logic. Actual simulation demonstrates camera-based discovery and confirmation, energy telemetry, charging, safe detours and reassignment after rover failure. Retained configurations and measurements make the implementation reviewable and policy comparison repeatable. Both policies complete all eight predefined layouts. In this batch the energy heuristic reduces mean distance by 2.90 m and discharge by 0.102 Wh, with varying timing effects between scenarios. These descriptive results concern the controlled simulation setting and its explicit models, rather than general superiority.

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